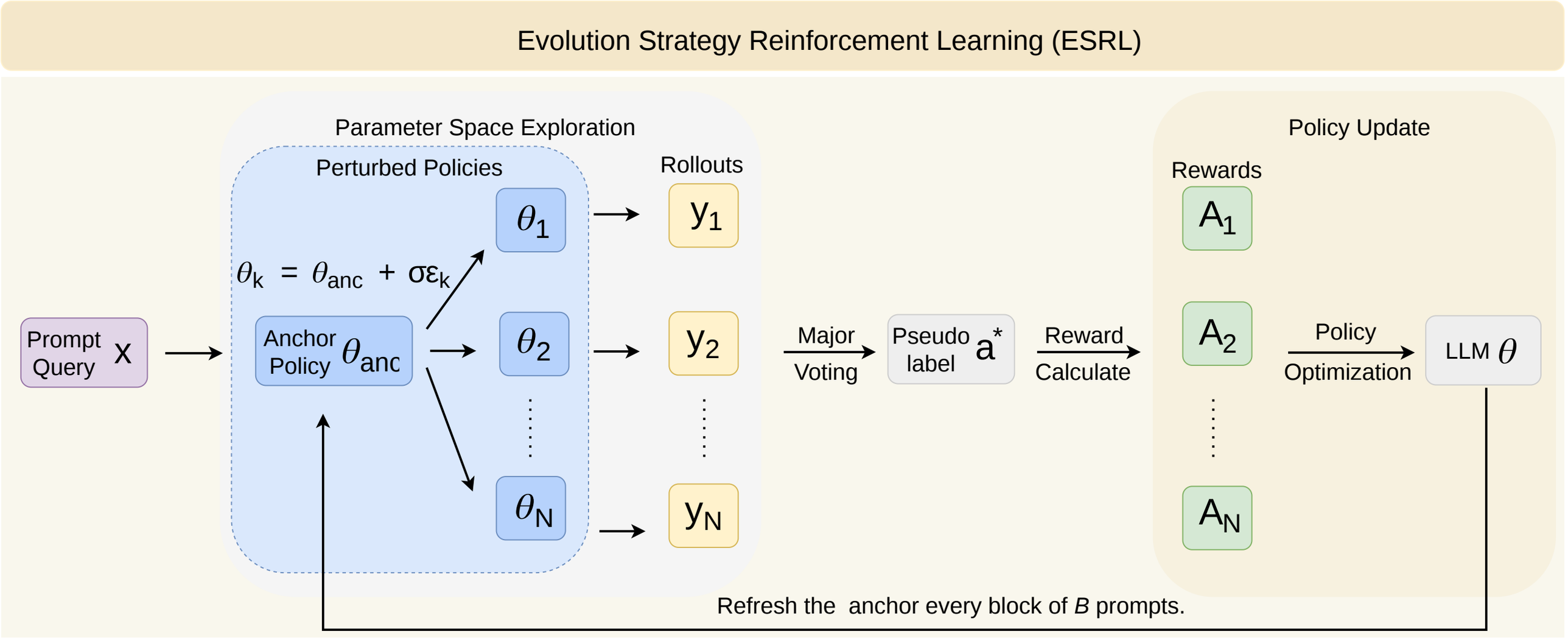
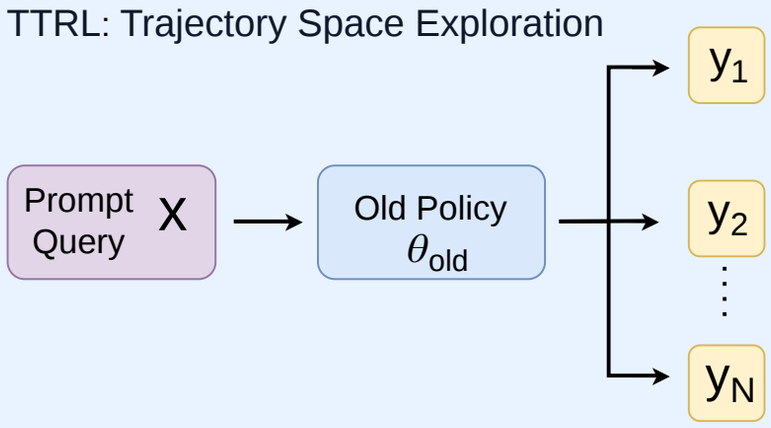
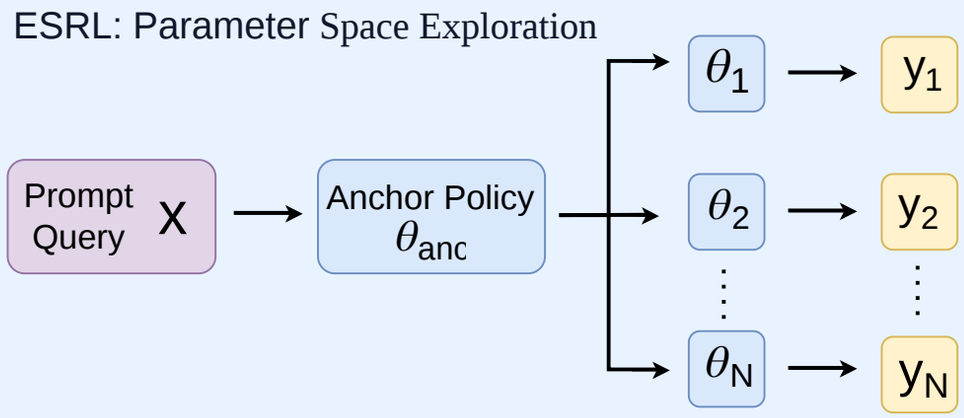


TTRL VS. ESRL Exploration Space





Policy Optimization Comparison

TTRL Objective

$$L_{\text{TTRL}}(\theta) = -\frac{1}{N}\sum_{k=1}^{N}\frac{1}{|y_k|}\sum_{t=1}^{|y_k|}\min\left(\varrho_{k,t}A_k,\ \text{clip}\left(\varrho_{k,t}, 1-\epsilon, 1+\epsilon\right)A_k\right) + \beta\,\widehat{\text{KL}},$$

$$\varrho_{k,t} = \frac{\pi_\theta(y_{k,t} \mid x, y_{k,<t})}{\pi_{\theta_{\text{old}}}(y_{k,t} \mid x, y_{k,<t})}.$$

ESRL Objective

$$L_{\text{ESRL}}(\theta) = \frac{1}{N}\sum_{k=1}^{N}\left[-A_k\frac{1}{|y_k|}\sum_{t=1}^{|y_k|}\log\pi_\theta(y_{k,t} \mid x, y_{k,<t}) + \lambda\widehat{\text{KL}}_k\right].$$